Unit 6, Lesson 7

Multiply Three- and Four-Digit Numbers by One-Digit Numbers

Standards

Addressing **4.NBT.B.5**

Instructional Routines

- Estimation Exploration

Goals

- Interpret (orally and in writing) diagrams that represent a partial product as a strategy for multiplying three- and four-digit numbers by place.
- Represent products of three- or four-digit and one-digit numbers with a diagram and calculate the value.

Student Facing Learning Goals

Let's multiply three- and four-digit numbers by one-digit numbers.

Lesson Purpose

The purpose of this lesson is for students to multiply a whole number of up to four digits by a one-digit number by decomposing factors by place value, finding partial products, and using properties of operations.

Narrative

In the previous lesson, students represented multiplication using base-ten diagrams and rectangular diagrams, and used place value reasoning to multiply two-digit numbers by one-digit numbers. In this lesson, they use rectangular diagrams and expressions to multiply up to four-digit numbers by one-digit numbers. They continue to use place value reasoning to decompose the multi-digit factor and to use partial products in their computation.

Students should have multiple opportunities to hear the term "partial products" referring to the results of multiplying a part of one factor and the other factor (or a part of one factor and a part of the other factor).

Access for Students with Disabilities

- Representation

Access for English Learners

- MLR2

Lesson Timeline

Warm-up	10 min
Activity 1	15 min
Activity 2	20 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

How can you leverage each of your student's ideas to support them in being seen and heard in tomorrow's math class?

Warm-up

10 min

Estimation Exploration: Mysterious Area

Standards

Addressing **4.NBT.B.5**

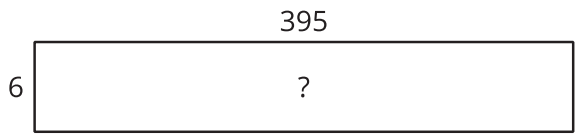
Instructional Routines

- Estimation Exploration

The purpose of an *Estimation Exploration* is to practice the skill of estimating a reasonable answer based on experience and known information. Listen for the different ways students use their understanding of place value to estimate the area and explain why an estimate is too low or too high.

Student Task Statement

What is the area of the rectangle represented by the diagram?



Record an estimate that is:

too low	about right	too high

Launch

- Groups of 2
- Display the image.
- "What is an estimate that's too high? Too low? About right?"
- 1 minute: quiet think time

Activity

- "Discuss your thinking with your partner."
- 1 minute: partner discussion
- Record responses.

Student Response

- Too low: 1,800–2,000.
- About right: 2,200–2,400
- Too high: 2,500 or higher

Activity Synthesis

- "Is your estimate greater or less than the actual product?" (greater)
- "How could we find the actual product?" ($6 \times 400 - 6 \times 5$, or subtract 6×5 from 2,400)

Activity 1

15 min

Larger Numbers to Multiply

Standards

Addressing **4.NBT.B.5**

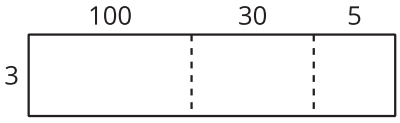
In this activity, students use rectangular diagrams to represent multiplication of three-digit and one-digit numbers. Though students may decompose the multi-digit factor in different ways, the activity is designed to encourage them to decompose it by place value—into hundreds, tens, and ones.

Access for English Language Learners

MLR2 Collect and Display. Synthesis: Direct attention to words collected and displayed from the previous activity. Invite students to borrow language from the display as needed, and update it throughout the lesson.
Advances: Conversing, Reading

Student Task Statement

1. Clare draws this diagram.



   a. What multiplication expression can be represented by the diagram?

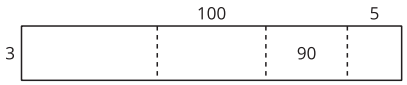
   b. Find the value of the expression. Show your thinking using diagrams, symbols, or other representations.

2. Consider the expression 6×252.

   a. Draw a diagram to represent the expression.

   b. Find the value of the expression. Show your thinking using diagrams, symbols, or other representations.

3. Lin draws a diagram to represent $3 \times 2{,}135$.



   a. Complete Lin's diagram.

   b. Write an expression to represent the value of each part of the diagram.

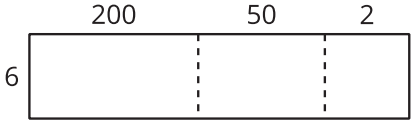
   c. Find the value of $3 \times 2{,}135$. Show your thinking using diagrams, symbols, or other representations.

Student Response

1. a. 3×135

   b. $(3 \times 100) + (3 \times 30) + (3 \times 5) = 300 + 90 + 15 = 405$

2. a.



Launch

- Groups of 2

Activity

- "Work with your partner on the first problem. Then try the rest of the activity independently."
- 2 minutes: group work time on the first problem
- 5–7 minutes: independent work time on the rest of the activity
- 3 minutes: partner discussion
- Monitor for students who:
  - Write expressions that show the multi-digit factor decomposed by place value.
  - Draw diagrams that partition the multi-digit factor by place value.

Activity Synthesis

- Invite several students to share diagrams they drew to represent 6×252.
- As each student shares, consider asking the class:
  - "Where do you see the parts of the factor that were decomposed?"
  - "What expressions are represented in the diagram?"
- Record expressions as students share.

b. $6 \times 200 = 1{,}200$
$6 \times 50 = 300$
$6 \times 2 = 12$
$1{,}200 + 300 + 12 = 1{,}512$

3. a.

	2,000	100	30	5
3	6,000	300	90	15

b. $(3 \times 2{,}000) + (3 \times 100) + (3 \times 30) + (3 \times 5)$

c. $3 \times 2{,}000 = 6{,}000$
$3 \times 100 = 300$
$3 \times 30 = 90$
$3 \times 5 = 15$
$6{,}000 + 300 + 90 + 15 = 6{,}405$

Activity 2

20 min

Jada's Errors

Standards

Addressing **4.NBT.B.5**

This activity extends students' work with multiplication to include a factor with up to four digits. Students begin to generalize that they could decompose any number into parts and multiply the parts. In this activity, students analyze a common error when multiplying. The work they look at does not apply place value understanding and therefore represents a product that is unreasonable for the given expression. When students analyze Jada's work, find her errors and explain their reasoning, they critique the reasoning of others (MP3).

Students see partial products as a way to describe the sub-products when a factor is decomposed and multiplied using the distributive property. Continue to refer to partial products in students' diagrams and calculations by their mathematical name to build students' intuition for their meaning (though students are not expected to use them in their reasoning).

Access for Students with Disabilities

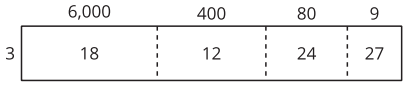
Representation: Access for Perception. Use base-ten blocks to demonstrate Jada's error and how her work should be corrected. Invite students to discuss how they might avoid such an error (for example, by estimating the product first, or by visualizing the value in base-ten blocks).
Supports accessibility for: Conceptual Processing, Visual-Spatial Processing

Student Task Statement

1. Jada uses this diagram to multiply $3 \times 6{,}489$. She makes a few errors in the diagram.

Launch

- Groups of 2



a. Explain the errors Jada made.

b. Find the value of $3 \times 6{,}489$. Show your thinking using diagrams, symbols, or other representations.

2. Find the value of 5×699. Show your thinking using diagrams, symbols, or other representations.

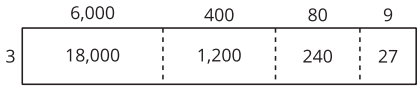
3. Find the value of $8 \times 4{,}973$. Show your thinking using diagrams, symbols, or other representations.

Student Response

1. Sample response:

   a. Jada forgot to include the place values (1,000, 100, and 10) when multiplying each part. She only multiplied the non-zero digits.

   b. 19,467



18,000 + 1,200 + 240 + 27 = 19,467

2. 3,495. Sample response:

$$\begin{aligned} 5 \times 699 &= (5 \times 600) + (5 \times 90) + (5 \times 9) \\ &= 3{,}000 + 450 + 45 \\ &= 3{,}495 \end{aligned}$$

3. 39,784. Sample response:

$$\begin{aligned} 8 \times 4{,}973 &= (8 \times 4{,}000) + (8 \times 900) + (8 \times 70) + (8 \times 3) \\ &= 32{,}000 + 7{,}200 + 560 + 24 \\ &= 39{,}784 \end{aligned}$$

Activity

- "Take a few quiet minutes to analyze Jada's errors. Then share your thinking with your partner."
- 2 minutes: independent work time on the first problem about Jada's diagram
- 2 minutes: partner discussion
- Pause for a discussion.
- "What did Jada do correctly?" (She multiplied the non-zero digits correctly. She decomposed the 6,489 by place value, which is helpful.)
- "What did Jada miss?" (She multiplied only the non-zero digits. She didn't account for the place value of the digits being multiplied. For example: The partial product of $3 \times 6{,}000$ is 18,000, not 18.)
- Display and correct Jada's diagram as a class.
- "Now complete the remaining problems independently."
- 5 minutes: independent work time on the last two problems
- Monitor for students who:
  - Draw diagrams that partition the multi-digit factor by place value.
  - Write expressions that show the multi-digit factor decomposed by place value.

Activity Synthesis

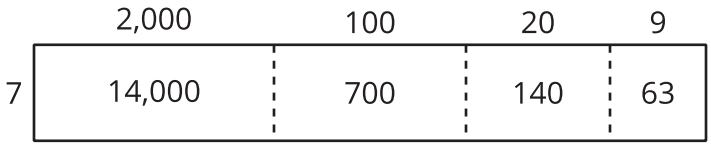
- Select 1–2 students to share their solutions and reasoning for finding the value of $8 \times 4{,}973$.
- Consider asking students to write expressions that represent the way they found each product. For example:

  $8 \times 4{,}973$
  $8 \times (4{,}000 + 900 + 70 + 3)$
  $(8 \times 4{,}000) + 8 \times 900 + 8 \times 70 + 8 \times 3$
  $32{,}000 + 7{,}200 + 560 + 24$
  $39{,}784$

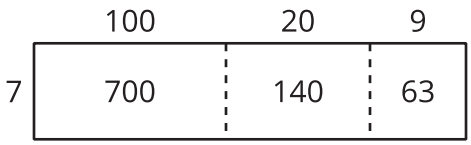
Lesson Synthesis

"Today we used diagrams to multiply three- and four-digit numbers by one-digit numbers. Let's compare diagrams that represent $7 \times 2{,}129$ and 7×129."

Display $7 \times 2{,}129$ and 7×129.



$14{,}000 + 700 + 140 + 63 = 14{,}903$



$700 + 140 + 63 = 903$

"How are the representations alike and how are they different?" (They both show the expanded form of the multi-digit factor, have 7 as one of the factors, and show partial products. They both show one factor decomposed by place value or written in expanded form. They are different because one diagram is decomposed into four parts because the factor 2,129 has four digits. The diagram shows 4 partial products. The other diagram is decomposed into 3 parts, because 129 has three digits. The diagram shows 3 partial products.)

"How would you find the value of $7 \times 2{,}039$?" (Think of 2,039 as $2{,}000 + 30 + 9$ and find $(7 \times 2{,}000) + (7 \times 30) + (7 \times 9)$.)

Suggested Centers

- Number Puzzles: Multiplication (4–5) , Stage 1: Two-Digit Factors (Addressing)
- Compare (1–5), Stage 3: Multiply within 100 (Supporting)

Cool-down

5 min

The Value of the Product

Standards

Addressing **4.NBT.B.5**

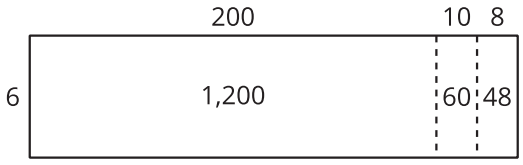
Student Task Statement

Find the value of 6×218. Show your thinking using diagrams, symbols, or other representations.

Student Response

1,308. Sample response:

$(6 \times 200) + (6 \times 10) + (6 \times 8)$
$= 1{,}200 + 60 + 48$
$= 1{,}308$



Responding to Student Thinking

Students may find the correct partial products but add them incorrectly.

Next Day Supports
Before the *Warm-up*, invite students to work in small groups to discuss a correct response to this *Cool-down*.